Project Management

The art and science of achieving results

Project Management Standards/Methodologies

- The PMBOK® Guide (Project Management Body of Knowledge)
- PRINCE2 (PRojects IN Controlled Environments)
- ISO 21502_2020
- Agile and Scrum
- Kanban

PMBOK® Guide – 7th Edition (Introduction)

What is PMBOK?

- PMI’s global standard for project management
- Provides a framework of principles and domains, not prescriptive processes
- Focuses on value delivery and adaptability

Key Elements in PMBOK 7th:

- 12 Principles of project management (e.g., stewardship, team, stakeholders, adaptability)
- 8 Performance Domains (e.g., Team, Stakeholder, Delivery, Planning, Measurement, etc.)
- Models, Methods, and Artifacts as supporting tools

Why Important for Students?

- Provides a universal language for projects
- Builds foundation for industry certifications (e.g., CAPM, PMP)
- Prepares you for both traditional and agile projects

What is a project?

*“A **temporary** endeavor undertaken to create a **unique** product, service, or result.*

The temporary nature of projects indicates a beginning and an end to the project work or a phase of the project work. Projects can stand alone or be part of a program or portfolio.”

PMBOK Guide 7th Edition

What is Project Management?

“The <u>application of knowledge, skills, tools, and techniques</u> to project activities to meet <u>project requirements</u>.

Project management refers to guiding the project work to deliver the intended outcomes.”

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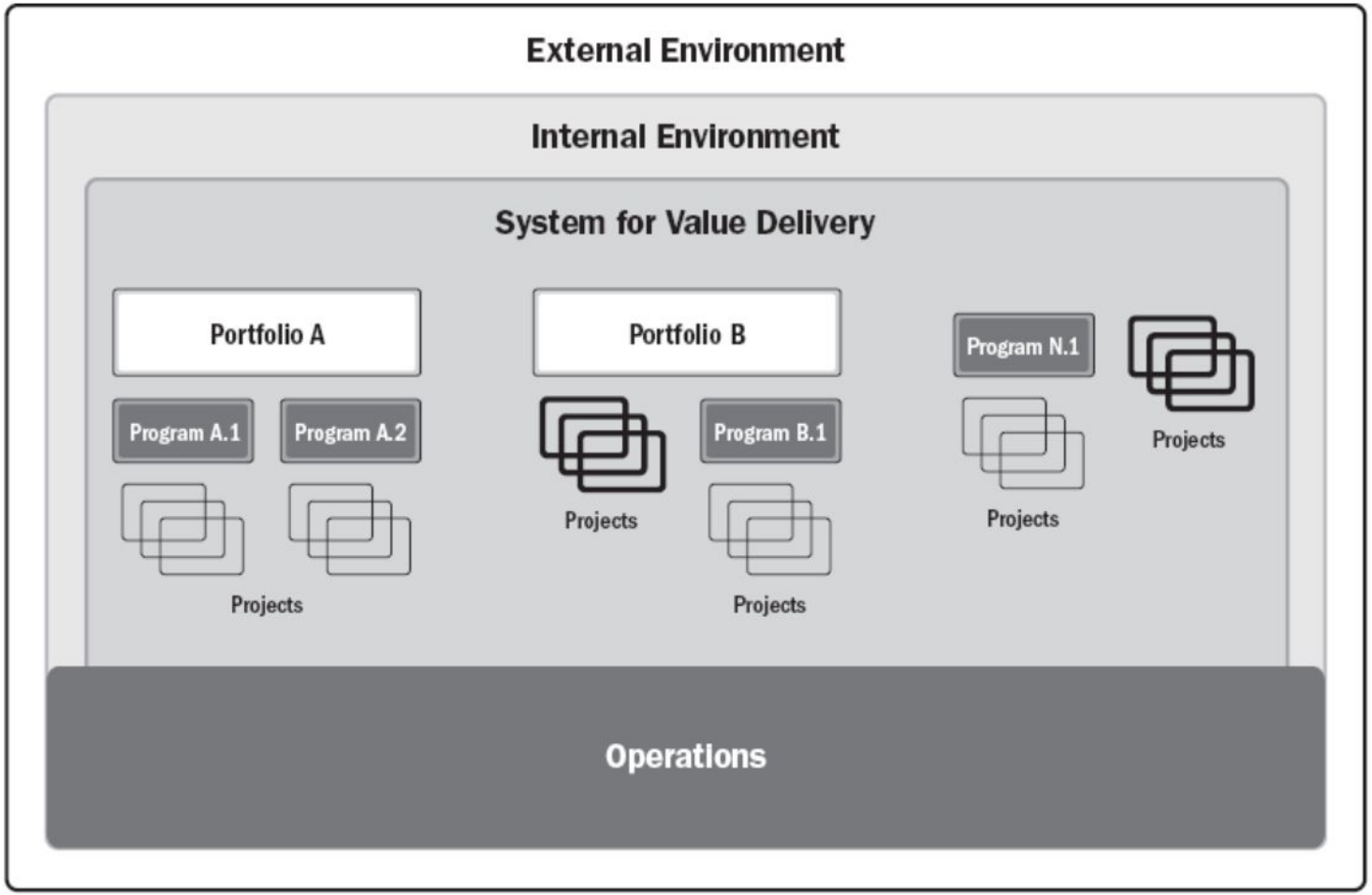

Figure 2-2. Components of a Sample System for Value Delivery

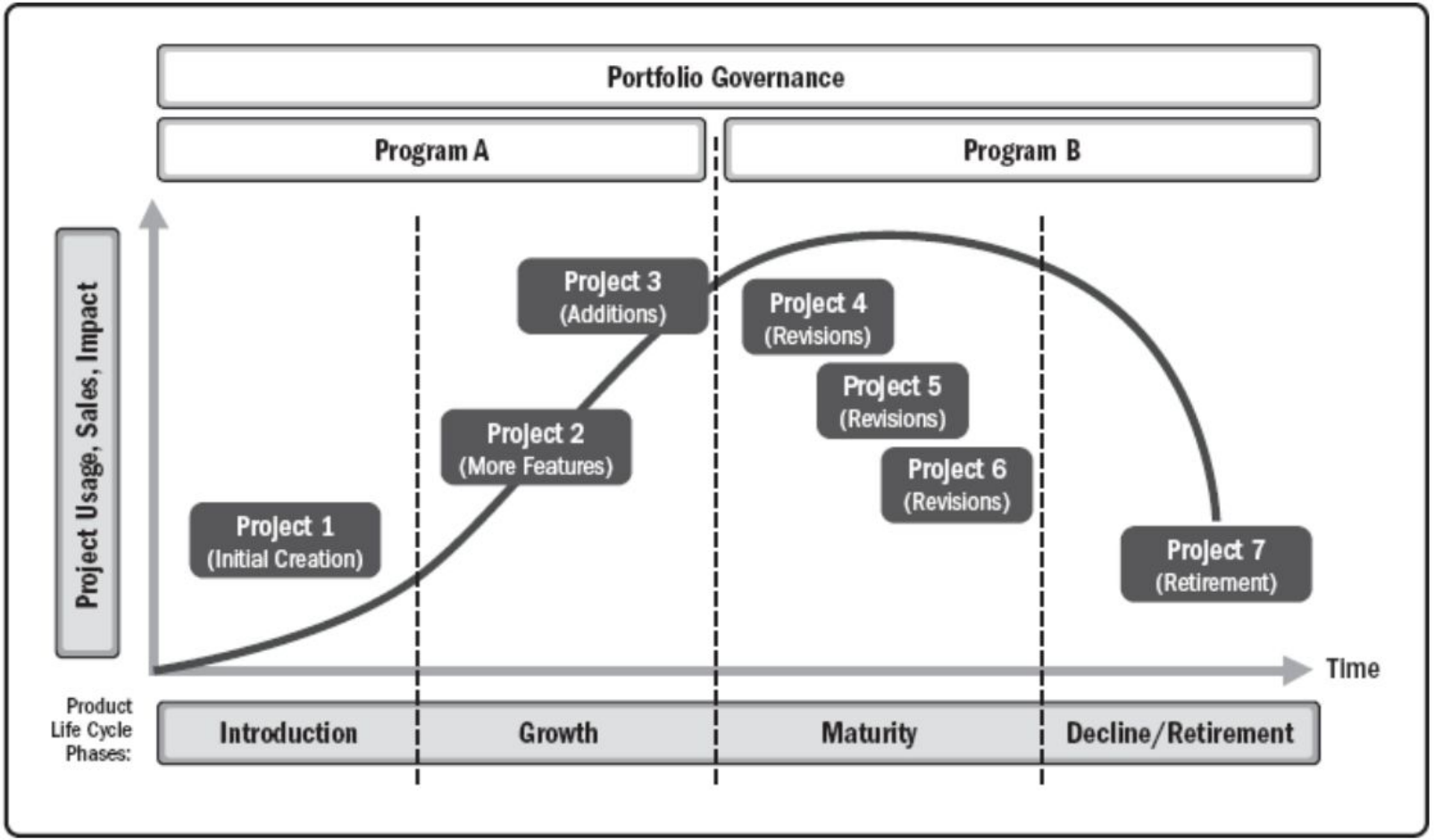

Figure 2-4. Sample Product Life Cycle

Value delivery and measurement

Projects do not simply produce outputs, but more importantly, enable those outputs to drive outcomes that ultimately deliver value to the organization and its stakeholders.

Evaluate effective performance through outcomes-focused measures, rather than through adherence to processes or the production of artifacts, plans, etc.

Project Management Principles

Project management principles serve as **guiding values** that help teams and leaders navigate projects in a structured yet flexible way. They ensure that projects deliver value, adapt to change, and are managed ethically and effectively.

- Provide a **universal foundation** for managing projects across industries.
- Help teams **make consistent, value-driven decisions** in complex environments.
- Ensure focus on **people, stakeholders, and collaboration**, not just tasks.
- Support **adaptability and resilience** in uncertain or changing conditions.

Project Management Principles

1. Be a diligent, respectful, and caring steward
2. Create a collaborative project team environment
3. Effectively engage with stakeholders
4. Focus on value
5. Recognize, evaluate, and respond to system interactions
6. Demonstrate leadership behaviors
7. Tailor based on context
8. Build quality into processes and deliverables
9. Navigate complexity
10. Optimize risk responses
11. Embrace adaptability and resiliency
12. Enable change to achieve the envisioned future state

The PMBOK® Guide

1. Be a diligent, respectful, and caring steward

Includes

1. Integrity
2. Care
3. Trustworthiness
4. Compliance

Both with internal and external point of view

2. Create a collaborative project team environment

Team agreements:

behavioral parameters and working norms established by the project team

Organizational structures:

Definitions of roles and responsibilities

Authority. The right to make decisions

Accountability. Answerable for an outcome. Accountability is not shared.

Responsibility. Being obligated to do or fulfill something. Responsibility can be shared.

3. Effectively engage with stakeholders

Stakeholders are

- individuals, groups, or organizations
- that may affect, be affected by,
- or perceive themselves to be affected by
- the project.

Identifying, analyzing, and proactively engaging with stakeholders

4. Focus on value

Value is the ultimate success indicator and driver of projects. e.g.

- Financial contribution
- Public good
- Customer’s perceived benefit

Value to

- the customer,
- the performing organization, or
- other stakeholders

Shift focus from deliverables to intended outcomes

5. Recognize, evaluate, and respond to system interactions

System thinking

- interconnections - internal and external
- Business case
- Big picture
- Integration of components, understanding, outcomes

Benefits

- Early uncovering of uncertainty/risks
- Better alignment
- Identify Opportunities

6. Demonstrate leadership behaviors

Create a project environment that prioritize

- vision,
- creativity,
- motivation,
- enthusiasm,
- encouragement, and
- empathy

7. Tailor based on context

- Each project is unique
- Adapt methods and procedures

- Processes (e.g. Develop a schedule)
- Methods (e.g. Critical Path Method)
- Artifacts (e.g. Gantt Chart)

8. Build quality into processes and deliverables

- Performance
- Conformity
- Reliability
- Resilience
- Satisfaction
- Efficiency

Focus on quality management processes

9. Navigate complexity

Complexity is the result of human behaviour, system interactions uncertainty and ambiguity

Project teams can stay vigilant in identifying elements of complexity and use a variety of methods to reduce the amount or impact of complexity.

Methods: e.g.

- Risk based prototyping
- Progressive elaboration

10. Optimize risk responses

- Individual and overall risks can impact projects.
- Risks can be positive (opportunities) or negative (threats).
- Risks are addressed continually throughout the project.
- An organization's risk attitude, appetite, and threshold influence how risk is addressed.

Risk responses should be:

- Appropriate for the significance of the risk,
- Cost effective,
- Realistic within the project context,
- Agreed to by relevant stakeholders, and
- Owned by a responsible person.

11. Embrace adaptability and resiliency

- Adaptability is the ability to respond to changing conditions.
- Resiliency is the ability to absorb impacts and to recover quickly from a setback or failure.
- A focus on outcomes rather than outputs facilitates adaptability.

- Short feedback loops to adapt quickly;
- Continuous learning and improvement;
- Regular inspection and adaptation of project work
- Open and transparent planning
- Deferring decision making to the last responsible moment;
- When an unplanned event occurs, project teams should look for potential positive outcomes that might be gained.

12. Enable change to achieve the envisioned future state

- Continually evaluating offerings for the benefit of stakeholders,
- rapidly responding to changes, and
- acting as agents for change.

- Helping stakeholders adapt to and adopt the new solution.
- Addressing resistance, uncertainty, and ambiguity.
- Building readiness and capability to operate successfully in the new environment.

Performance Domains

Performance domains in PMBOK 7th Edition describe broad areas of focus that are critical for delivering successful project outcomes. Unlike rigid processes, they provide flexibility and emphasize principles, behaviors, and activities that help teams adapt to different project contexts.

- Ensure **focus on value delivery** rather than just outputs.
- Provide a **holistic view** of what drives project success.
- Guide **team behavior, stakeholder engagement, and planning**.
- Support **tailoring and adaptability** across predictive, agile, and hybrid projects.

The PMBOK® Guide

1. Stakeholder
2. Team
3. Development Approach and Life cycle
4. Planning
5. Project Work
6. Delivery
7. Measurement
8. Uncertainty

Project Performance Domains

1. Stakeholder

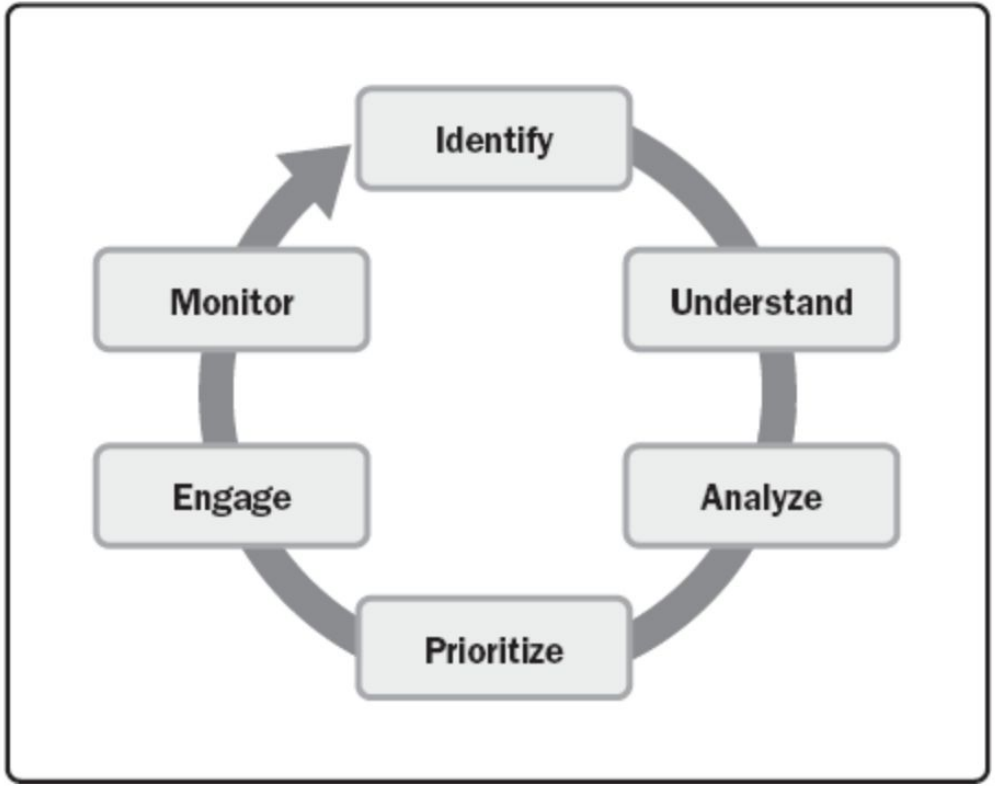

Figure 2-3. Navigating Effective Stakeholder Engagement

2. Team

- Project team management and leadership
  - Centralised vs distributed management
- Project team Culture
  - Transparency
  - Integrity
  - Respect
  - Courage
  - Support
- Leadership skills
  - Critical thinking
  - Motivation
  - Emotional intelligence
  - Conflict management

3. Development Approach and Life Cycle

- Development Approach
  - Predictive
  - Iterative
  - Incremental
- Selecting development approach
  - Innovation
  - Requirements
  - Risk
  - Safety requirements
  - Regulations
- Delivery Cadence
  - Single delivery
  - Multiple deliveries
  - Periodic deliveries
  - Continuous deliveries

3. Development Approach and Life Cycle

- Life cycle
  - Phases

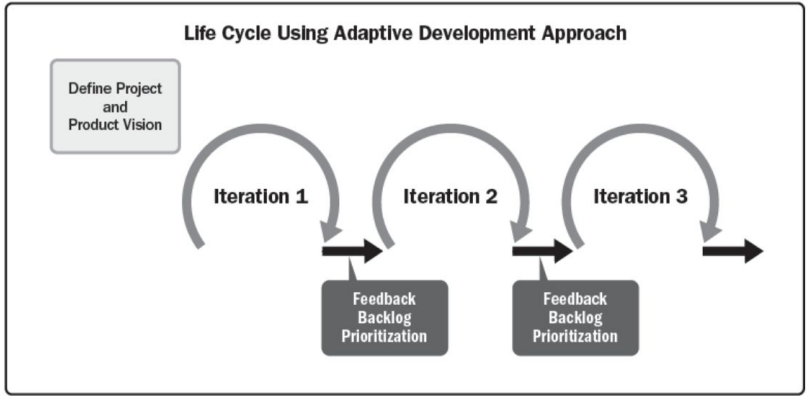

Figure 2-11. Life Cycle with Adaptive Development Approach

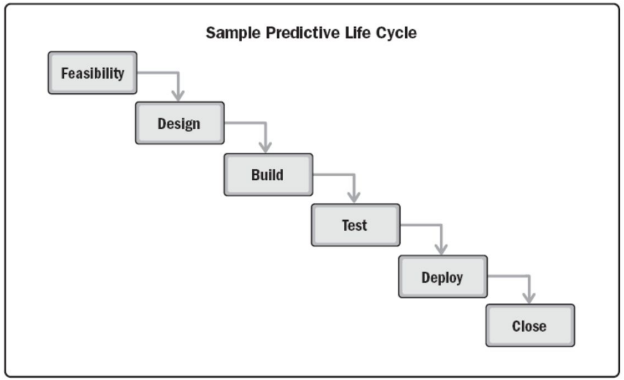

Figure 2-9. Sample Predictive Life Cycle

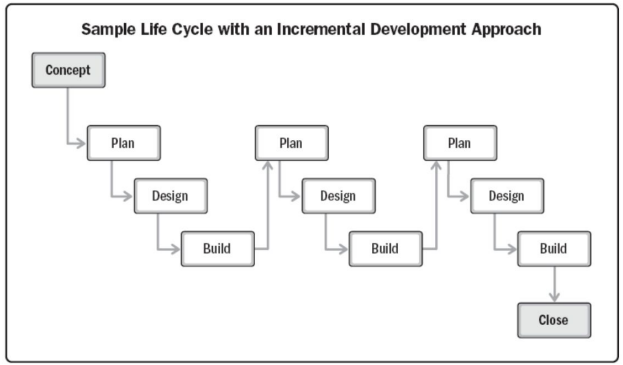

Figure 2-10. Life Cycle with an Incremental Development Approach

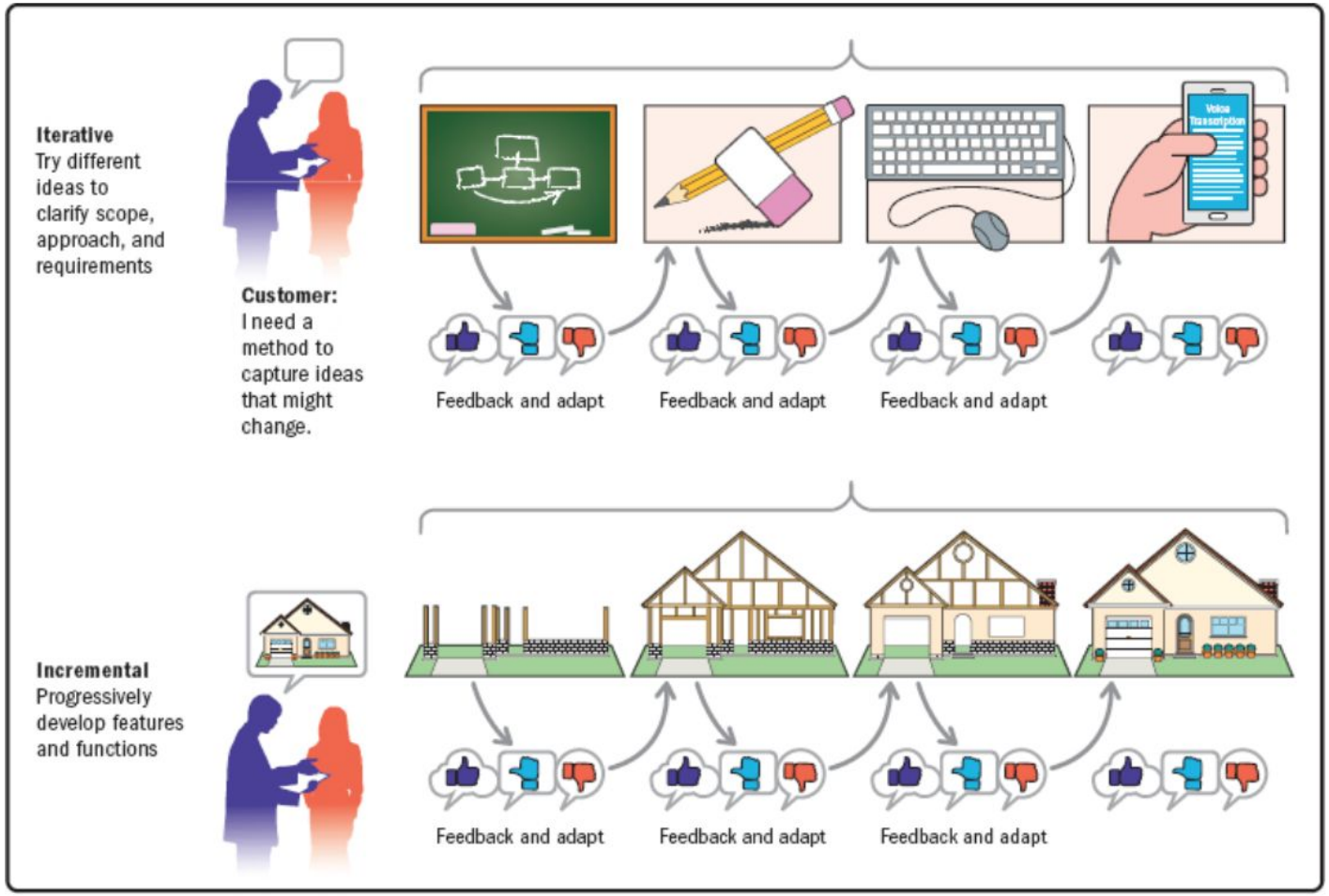

Figure 2-8. Iterative and Incremental Development

4. Planning

- Product scope
- Project scope
- Estimating
- Scheduling
- Budget
- Team composition
- Communication plan
- Procurement
- Change management
- Measurement

5. Project Work

- Establishing the processes
- Performing the work

- Executing everything planning in the Panning performance domain

- Knowledge management

6. Delivery

- Delivery of value
- Requirements
- Scope definition
- Completion
- Quality

7. Measurement

- A reliable understanding of the status of the project.
- Actionable data to facilitate decision making.

- Key performance indicators (KPIs)
  - Leading indicators. Leading indicators predict changes or trends in the project.
    - Size of the project (Function points, story points etc.)
    - WIP (number of items in progress)
    - Availability of stakeholders
    - Clarity of success criteria
  - Lagging indicators. Lagging indicators measure project deliverables or events. They provide information after the fact.
- Metrics
  - Defects
  - Performance
  - Work in progress
  - Schedule variance
  - Cost variance
  - Burn charts (e.g. based on story points)

8. Uncertainty

- Successfully navigating uncertainty begins with understanding the larger environment within which the project is operating.

- Ambiguity
  - Progressive elaboration
  - Experiments
  - Prototypes
- Complexity
  - Decomposition
  - Simulation
- Risks
- Contingency reserve
- Management reserve

Models, Methods, and Artifacts

Model. A model is a thinking strategy to explain a process, framework, or phenomenon.

Method. A method is the means for achieving an outcome, output, result, or project deliverable.

Artifact. An artifact can be a template, document, output, or project deliverable.

References

- PMBOK® Guide – 7th Edition